Aposteriori Unimodality: Attributing polarization to sociodemographic groups

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March 2025

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields such as content moderation, toxicity detection and hate speech detection. These tasks are especially essential for training systems that protect vulnerable and minority groups in online spaces, where they are often targeted [17, 4]. However, annotations for such tasks are subjective, and are usually based on majority-group annotators. This may limit the usefulness of the data, or even lead to biased and misconfigured systems.

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree in vain. An example of such cases would be racist comments that are presented with coded language (usually referred to as "dog-whistles" [13]), which are not picked up by most annotators, but only by ones belonging in the targeted minority. Inversely, majority-group annotators may bias their own annotations against the minority-group annotators; CITATION for instance shows that toxicity models disproportionally marked African American speech as "toxic". This issue exists even in representative samples.

Inter-annotator agreement is often used to detect such cases [8]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [11, 12]—specifically whether two or more annotation clusters are present. Figure 1 displays how the two methods handle four common cases of annotation distributions for a hypothetical toxicity annotation task.

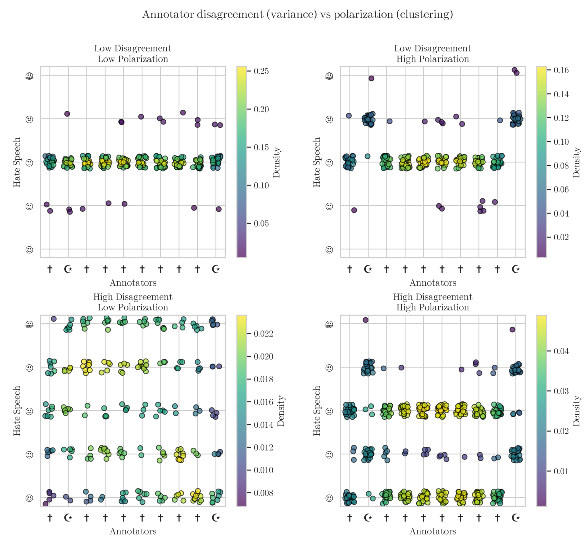

Figure 1: Disagreement is similar to variance; thus it overlooks cases where annotators may be "split down the middle". Polarization on the other hand, is able to identify multiple clusters, even if they are close together.

While we can detect polarization [12, 11], there is little work on how to detect whether it is actually caused by marginalized groups. Pavlopoulos and Likas [12] propose a mechanism which only works in the extreme case where overall polarization is zero; in this work we demonstrate that this case is not frequent in real-world data. We instead propose the *"Apunim Metric"*, which attributes polarization to individual annotator sociodemographic groups. We provide a formulation which avoids inherent statistical limitations. Our contributions are:

1. We introduce a new metric which attributes polarization to specific sociodemographic groups.

2. We discover issues not yet acknowledged in literature, such as the presence of inherent polarization and conflicting polarization directions. Our proposed metric is designed to avoid these issues.

3. We apply our test to four datasets; two with human comments and annotations [15, 9] as well as two generated and annotated by Large Language Model (LLM) agents [16].

2 Methodology

In this section, we provide a formal mathematical formulation for the problem of attributing polarization to specific annotator characteristics (§2.1), and offer an intuitive rationale for how established polarization metrics can be leveraged (§2.2). We then introduce a data-point-level statistic that attributes polarization to individual SocioDemographic Background (SDB) groups (§2.3), and subsequently develop a statistic that generalizes this mechanism to an entire dataset.

2.1 Problem Formulation

SocioDemographic Backgrounds Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to isolate individual groups within a SDB. Let Θ be one of multiple "dimensions" the set of all annotator SDB groups for a single dimension (e.g. "male", "female" and "non-binary" if we are investigating annotator gender).

Annotations Now let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated data points. We assume that annotating a data point depends on two variables: inherent ("apriori") controversy, and the ("aposteriori") annotator's SDB, as well as uncontrolled factors such as mood and personal experiences (noise). Assuming that each data-point c is assigned multiple annotators, we can define the annotation multi-set $A(c) = \{(a, \theta)\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task would be formulated as:

$$\begin{aligned}&A(\textit{"Could be better, could be worse"}) =\\&\{(\textit{positive}, \textit{female}), (\textit{negative}, \textit{male}),\\&(\textit{positive}, \textit{female}), \ldots\}\end{aligned} \tag{1}$$

Polarization Each set of annotations features a certain degree of *polarization.* Unpolarized annotations are usually unimodal; the annotators disagree on the details (which would be shown roughly as a bell curve around the median annotation), not fundamentally (which would likely be shown as a multimodal distribution). Pavlopoulos and Likas [12] create a polarization metric, the "normalized Distance From Unimodality (nDFU)", which measures whether an annotation set shows no polarization (unimodal distribution — $nDFU \to 0$), up to complete polarization (multimodal distribution — $nDFU \to 1$). Thus, we need a metric that measures how much the nDFU of a data point can be partially explained by θ (given that polarization exists).

2.2 Quantifying changes in polarization

Single data point polarization Intuitively, θ partially explains the polarization of a data point c when the annotations grouped by θ are more polarized compared to the full set of annotations. Figure 3 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators.[1] The annotations are generally polarized ($nDFU_{all} = 0.625$). The annotations by female annotators exhibit low polarization between themselves ($nDFU_{women} = 0.1$), since most agree the data-point

[1] The use of only two predominant genders is made only for the purposes of demonstration.

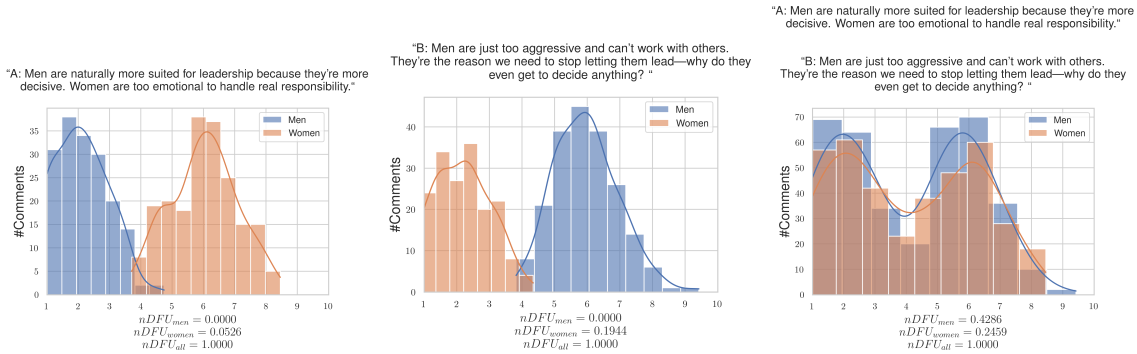

Figure 2: Hypothetical example of a polarizing discussion with two comments, for which the annotators disagree on which is the toxic one. If we aggregate the two comments, the polarization scores for both men and women significantly rise, obscuring whether the exhibited polarization can be partially attributed to gender.

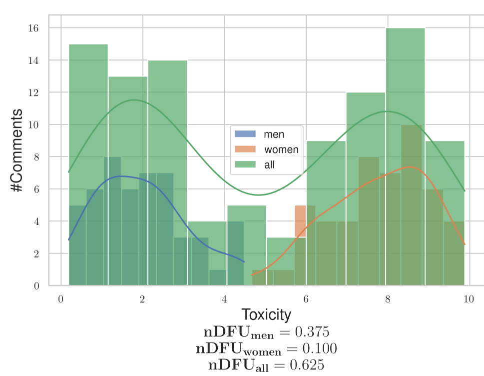

Figure 3: Hypothetical example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender. Overall polarization is much greater than the polarization exhibited by the annotations grouped by gender.

is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.3725$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by disagreements between male and female annotators.

Dataset polarization Given this observation, we would be tempted to aggregate all annotations in the dataset and compare the polarization of each θ with the full set of annotations $A(c)$. In fact this is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each data point may be annotated by only 5 annotators, leading to a best-case 3-2 split between groups.

However, this formulation would not work well. Figure 2 illustrates a hypothetical discussion with two comments, both of which are toxic, but where male and female annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This example demonstrates that polarization has a *direction*, which is not captured by the $nDFU$ metric which is symmetric. Thus, care must be taken to not combine polarization effects of opposite directions.

2.3 The pol-statistic

As demonstrated above, it is necessary to define a metric for the polarization attribution of each individual data point. We thus introduce the *"pol-*

statistic", which quantifies the change in nDFU when annotations are grouped by $\theta \in \Theta$.

In order to define this statistic, we need to measure (1) the polarization of a data-point when grouped by θ, and (2) the "apriori" polarization § 1 present in the annotations, which is driven by either the inherent controversy of the data-point. Note that we can not directly compare the θ-subset of annotations with the full set of annotations, since any statistical comparison between them violates the assumption of independent samples.

The observed polarization of a data-point on the group of annotations θ can be directly computed as:

$$pol(c, \theta) = nDFU(P(A(c), \theta) \tag{2}$$

where $P(A(c), \theta_i) = \{a(c; \theta) \in A(c) | \theta = \theta_i\}$ is the partition of A for a data-point c, for which its annotators belong to the SDB group θ_i.

We can estimate the apriori polarization in data point c by bootstrapping. We randomly partition the annotations in $|\Theta|$ groups, with matching group sizes (e.g., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we will create random partitions of sizes 80 and 20). The randomly partitioned annotations are then sampled t times. Formally, the expected inherent polarization will be given by:

$$\frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{3}$$

where $\tilde{P}_i$ is the random partition operator with matching group sizes.

2.4 Aposteriori Unimodality

By obtaining the pol-statistics for all data-points in a dataset d for SDB θ, we can get the mean aposteriori polarization for the dataset:

$$pol_{apost}(d, \theta) = \frac{1}{|d|} \sum_{c \in d} pol(c, \theta) \tag{4}$$

and similarly the mean apriori dataset polarization:

$$pol_{apriori}(d) = \frac{1}{|d|} \sum_{c \in d} \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{5}$$

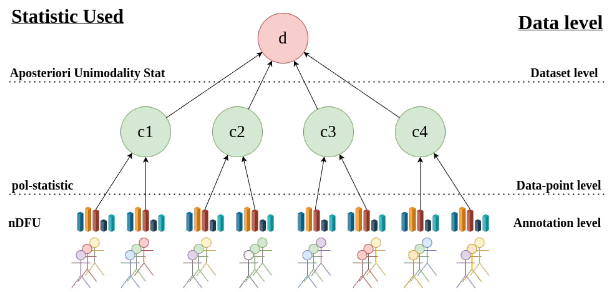

Figure 4: An overview of the Aposteriori Unimodality Test. We gather statistical information from the annotations (different SDBs are denoted by different colors) via the nDFU measure, which is aggregated on the data-point-level by the pol-statistic. The Aposteriori Unimodality Test is applied on the discussion level, operating on the pol-statistics of the individual data-points.

We can then define the *Apunim Statistic* as:

$$apunim(d, \theta) = \frac{pol_{apost}(d, \theta) - pol_{apriori}(d)}{1 - pol_{apriori}(d)} \tag{6}$$

If $apunim(d, \theta) \approx 0$, the exhibited polarization among annotators of group θ does not surpass what would be expected by chance. If $apunim(d, \theta) > 0$, then the group partially explains a rise in polarization, and inversely for the case $apunim(d, \theta) < 0$. The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 4.

While it is possible to obtain p-value results for the apunim values being larger or less than 0 respectively, we find that the inherent variance of our statistic makes a p-value estimation worthless ($p \gg 0.1$). Nevertheless, we include a definition for parametric and non-parametric p-value estimation, and our results in Appendix B.

2.5 Reliability of the pol statistic

In §2.4 we mentioned that the test relies on enough annotations for each SDB group and data-point. This can be an issue, given the cost required to utilize multiple human annotators for each data-point [14].

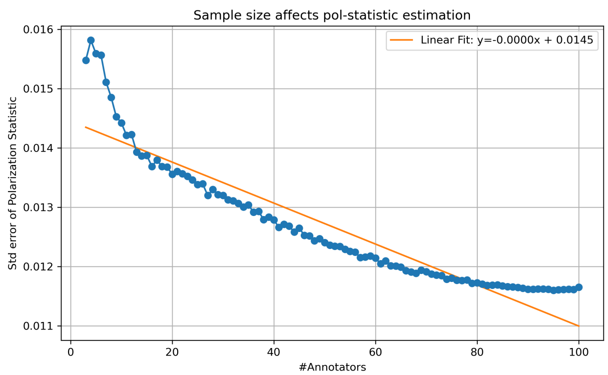

Figure 5: Standard error of the *pol-statistic* when sampled with replacement from the 100 LLM-annotator synthetic dataset for various number of annotators.

Table 1: Aposteriori Unimodality kappa and pvalue results for the Sap et al. 2022 dataset

SDB Feature		apunim
Age	(-0.001, 20.0]	0.000000
	(20.0, 40.0]	-0.028003
	(40.0, 60.0]	0.035342
	(60.0, 80.0]	0.078341
Ethnicity	black	-0.419412
	hisp	0.000000
	other	0.000000
	white	0.149386
Gender	man	0.151317
	nonBinary	0.000000
	woman	-0.349321

We use a subset of synthetic discussions from a synthetic dataset, the Virtual Moderation Dataset (VMD) [16]. We use 100 LLM annotators with different SDB characteristics and sample progressively 3-100 annotators with replacement, then calculate the standard error with the mean *pol-statistic*. As expected, the standard error is inversely proportional to the number of annotators, although the difference is not great, even for a low number of annotators.

Figure 5 demonstrates the effect of the number of annotators to the *pol-statistic* estimation. We can see that the pol statistic becomes consistent with about 15 annotators per comment. After about 80 annotators, we start to get diminishing results. This highlights a core limitation of the apunim metric.

3 Datasets

We use the datasets provided by Kumar et al. [9] and Sap et al. [15]. The datasets feature various online comments, each annotated by a number of human annotators (5 and 4-6 annotators per comment respectively). There is a reasonable amount of polarization in most annotations for both dataset, as demonstrated by Figure 6.

The Sap et al. [15] dataset includes racism annotations for 626 Twitter/X comments, and standard SDB information (age, race, education, gender). The Kumar et al. [9] dataset measures toxicity on various online comments, and the provided SDBs include mostly annotator experiences (e.g., whether they have been personally targeted online) as well as standard SDB information such as sexual orientation, age and education. Since the dataset is extensive (106035 comments), we only select a sample of 10000 comments, due both to computational constraints.

4 Results

Assuming a confidence level of $\alpha = 0.1$ we test whether any of the provided SDB dimensions can partially explain polarization in the toxicity/racism annotations. The results for the Kumar et al. [9] and Sap et al. [15] datasets can be found in Tables **??**, **??** respectively.

5 Conclusion

We introduced the "Aposteriori Unimodality Test", a statistical test that detects whether certain sociodemographic groups partly cause polarization in annotations. Our test is resistant to low samples of an-

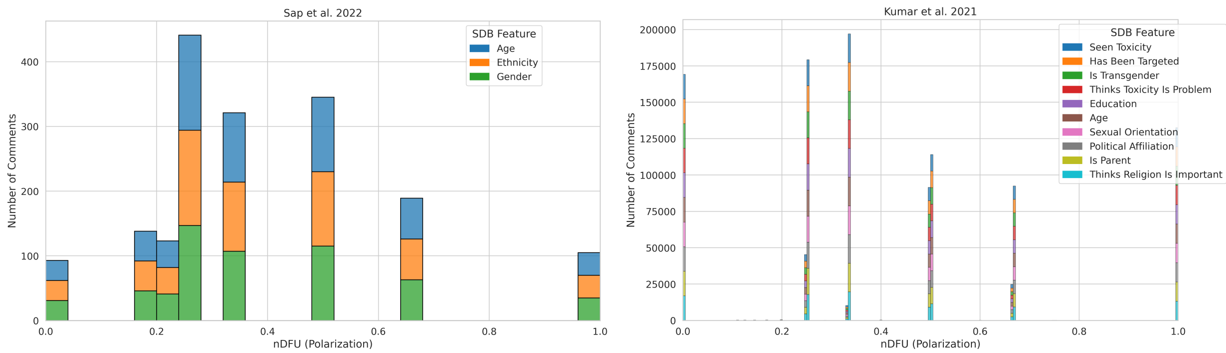

Figure 6: Histogram of dataset polarization for the human datasets considered in this paper per SDB feature.

notations, bypasses common statistical issues such as sample dependence, and avoids newly-discovered issues such as the presence of inherent polarization and polarization direction. We apply our test two human-generated and two LLM-generated datasets and find interesting patterns on the former, and verify current literature on LLM SDB prompting on the latter.

6 Limitations

Since there are no other established tests that attribute polarization to sociodemographic characteristics, there is no way to verify that the results presented in §4 are accurate. Similarly, there is no qualitative way of evaluating our test, other than observations on the (non-) efficacy of LLM SDB prompting and our own intuition. Lastly, while our test seems to work sufficiently well with a small number of annotators per data-point, we still encourage researchers using this test to have at least a few annotators of each sociodemographic group that is being tested, in their dataset.

7 Ethical Considerations

While our works aims to help protect marginalized and disadvantaged groups by attributing polarization to certain subgroups, it can be taken advantage of by malicious actors. Given a dataset with fine-grained SDB information, these actors can use our test to target specific vulnerable groups. We thus urge researchers to keep datasets including such information protected, and provided to others only under explicit permission.

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Table 2: Aposteriori Unimodality kappa and pvalue results for the Kumar et al. 2021 dataset

SDB Feature		apunim
Age	18 - 24	-0.030410
	25 - 34	0.040010
	35 - 44	-0.010025
	45 - 54	0.018776
	55 - 64	0.003176
	65 or older	0.017648
	Prefer not to say	0.000000
Education	Associate degree	-0.055727
	Bachelor's degree	0.095408
	College, no degree	-0.108984
	Doctoral degree	0.008115
	High School graduate	-0.040907
	Master's degree	0.054708
	No high school	-0.007941
	Other	0.000000
	Prefer not to say	0.000000
	Professional degree	0.001086
Targeted	False	-0.095286
	True	0.112036
Is Parent	No	-0.136264
	Prefer not to say	-0.004510
	Yes	0.093870
Is Transgender	No	-0.151712
	Prefer not to say	0.515608
	Yes	0.016393
Politics	Conservative	0.066361
	Independent	-0.016543
	Liberal	-0.039058
	Other	-0.018498
	Prefer not to say	-0.001853
Seen Toxicity	False	0.145958
	True	-0.117346
Sexual Orient.	Bisexual	0.179856
	Heterosexual	-0.131972
	Homosexual	-0.044238
	Other	0.003322
	Prefer not to say	-0.028099
	NaN	0.077491
Religion	Not important	-0.167580
	Not too important	-0.026466
	Prefer not to say	0.004367
	Somewhat important	0.036322
	Very important	0.088742
Toxicity	Frequently a problem	-0.057269
	Not a problem	0.039260
	Occasionally a problem	-0.009915
	Rarely a problem	0.044619
	Very frequently a problem	-0.057531

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A Acronyms

VMD Virtual Moderation Dataset

LLM Large Language Model

SDB SocioDemographic Background

nDFU normalized Distance From Unimodality

FWER Family-Wise Error Rate

B p-value Estimation

B.1 Methodology

Like most metrics, we also include a p-value alongside the *apunim* value. This can be computed either non-parametrically, by repeatedly shuffling annotations and computing a null distribution of *apunim*, or parametrically, by assuming a normal distribution (which is generally a safe assumption if $t \geq 30$). In the latter case, we compute:

$$z = \frac{\hat{\kappa} - \mathbb{E}[\kappa]}{SE(\hat{\kappa})}, p = 1 - \Phi(z) \tag{7}$$

Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values. We choose the Holms method [6]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [3]). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

B.2 Results

The results for the datasets presented in § 3 can be found in Tables **??**, **??** for the Sap et al. [15] and Kumar et al. [9] datasets respectively. The experiments were ran on the exact same parameters as in § 4.

Note the absence of any statistically significant results for both parametric and non-parametric tests. This suggests that, although the differences in polarization captured by the *pol* statistic introduced in

§ 2 are substantial enough to hint at polarization being attributable to certain groups, they remain too small to overcome the inherent noise in human annotations. As a result, we cannot obtain a robust estimate of statistical significance.

C Synthetic Datasets

We use two synthetic datasets; one is the VMD presented in Tsirmpas, Androutsopoulos, and Pavlopoulos [16]. This dataset features 140 discussions, each having 14 comments (and usually up to 14 facilitator comments), each comment annotated by 10 LLM annotators, each supplied with a different SDB. The second dataset, is an extension of VMD, where we select four random unmoderated discussions, and employ 100 distinct LLM annotators.

An issue with synthetic datasets is the lack of overall polarization (see Figure 7). This can be offset by the ability of synthetic datasets to be scaled to a much greater extent than human datasets. This is because LLMs do not have cost and availability constraints of humans.

The results for the 100-annotator dataset can be found in Table **??**. We can not obtain results for the VMD dataset, since no comment exhibited apriori polarization while also being represented by more than two annotators of different groups. The lack of overall polarization, as well as the quantitatively smaller apunim values, suggest that SDB prompting can not be used to simulate annotations for human groups. This is consistent with relevant literature on the topic [1, 5, 14, 7, 2, 10].

Table 3: Aposteriori Unimodality kappa and pvalue results for the Sap et al. 2022 dataset

SDB Feature		apunim	pvalue_parametric	pvalue_nonparametric
Age	(-0.001, 20.0]	0.000000	NaN	1.000000
	(20.0, 40.0]	-0.028003	0.946350	0.914439
	(40.0, 60.0]	0.035342	0.934080	0.930108
	(60.0, 80.0]	0.078341	0.757341	1.000000
Ethnicity	black	-0.419412	0.413176	0.457732
	hisp	0.000000	NaN	1.000000
	other	0.000000	NaN	1.000000
	white	0.149386	0.690118	0.713402
Gender	man	0.151317	0.711524	0.700272
	nonBinary	0.000000	NaN	1.000000
	woman	-0.349321	0.413578	0.501362

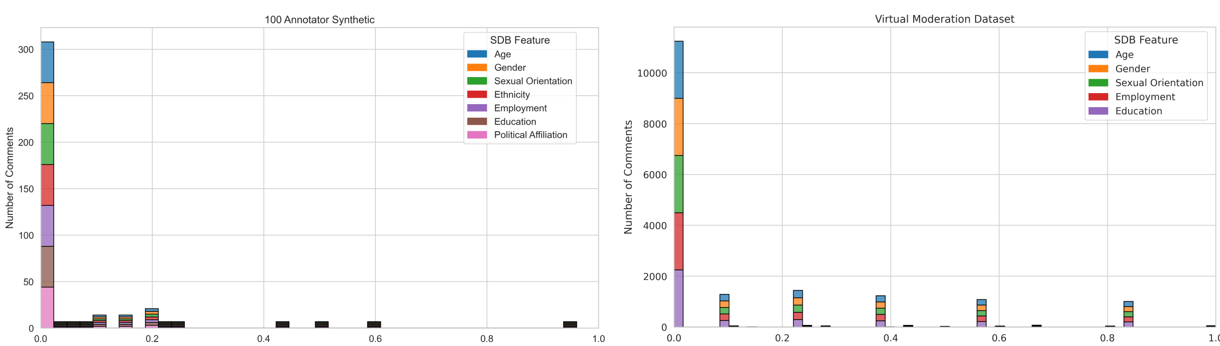

Figure 7: Histogram of dataset polarization for the synthetic datasets considered in this paper per SDB feature.

Table 4: Aposteriori Unimodality kappa and pvalue results for the Kumar et al. 2021 dataset

SDB Feature		apunim	pvalue_parametric	pvalue_nonparametric
Age	18 - 24	-0.030410	0.954856	0.967112
	25 - 34	0.040010	0.948614	0.963320
	35 - 44	-0.010025	0.987850	0.998125
	45 - 54	0.018776	0.973235	0.971674
	55 - 64	0.003176	0.994370	1.000000
	65 or older	0.017648	0.959945	1.000000
	Prefer not to say	0.000000	NaN	1.000000
Education	Associate degree	-0.055727	0.923705	0.947000
	Bachelor's degree	0.095408	0.878875	0.908507
	College, no degree	-0.108984	0.862214	0.927355
	Doctoral degree	0.008115	0.953037	1.000000
	High School graduate	-0.040907	0.937773	0.980510
	Master's degree	0.054708	0.935074	0.978080
	No high school	-0.007941	0.978720	1.000000
	Other	0.000000	NaN	1.000000
	Prefer not to say	0.000000	NaN	1.000000
	Professional degree	0.001086	0.994127	1.000000
Has Been Targeted	False	-0.095286	0.851124	0.840822
	True	0.112036	0.838550	0.851233
Is Parent	No	-0.136264	0.793274	0.834386
	Prefer not to say	-0.004510	0.983443	1.000000
	Yes	0.093870	0.851837	0.852657
Is Transgender	No	-0.151712	0.796024	0.776316
	Prefer not to say	0.515608	0.356331	0.416667
	Yes	0.016393	0.973112	0.969231
Political Affiliation	Conservative	0.066361	0.918828	0.974428
	Independent	-0.016543	0.979319	0.992056
	Liberal	-0.039058	0.951880	0.977920
	Other	-0.018498	0.949256	1.000000
	Prefer not to say	-0.001853	0.996053	1.000000
Seen Toxicity	False	0.145958	0.789892	0.810560
	True	-0.117346	0.816276	0.831252
Sexual Orientation	Bisexual	0.179856	0.786152	0.789989
	Heterosexual	-0.131972	0.808635	0.824752
	Homosexual	-0.044238	0.940537	0.940860
	Other	0.003322	0.979759	1.000000
	Prefer not to say	-0.028099	0.957889	1.000000
	NaN	0.077491	0.756128	1.000000
Thinks Religion Is Important	Not important	-0.167580	0.790175	0.858974
	Not too important	-0.026466	0.961419	0.982239
	Prefer not to say	0.004367	0.983731	1.000000
	Somewhat important	0.036322	0.954503	0.975619
	Very important	0.088742	0.893641	0.964399
Thinks Toxicity Is Problem	Frequently a problem	-0.057269	0.927206	0.983830
	Not a problem	0.039260	0.934240	0.997835
	Occasionally a problem	-0.009915	0.987592	0.996255
	Rarely a problem	0.044619	0.943330	0.956728
	Very frequently a problem	-0.057531	0.914521	0.954014

Table 5: Apunim results for the 100 dataset

SDB Feature		apunim	p_param	p_nonparam
Age	(14.927, 33.25]	-0.023	0.949	0.889
	(33.25, 51.5]	-0.004	0.998	0.944
	(51.5, 69.75]	0.022	0.929	0.833
	(69.75, 88.0]	0.024	0.922	0.889
Education	high-school	-0.034	0.921	0.889
	none	0.036	0.889	0.778
	university	-0.017	0.966	0.889
Employment	blue-collar	-0.002	0.995	0.944
	unemployed	-0.026	0.944	0.889
	white-collar	0.021	0.932	0.833
Ethnicity	asian	-0.036	0.916	0.722
	black	-0.024	0.948	0.889
	other	0.069	0.803	0.722
	white	-0.061	0.843	0.778
Gender	female	0.002	0.981	1.000
	male	0.030	0.909	0.833
	non-binary	-0.107	0.690	0.611
Political Affiliation	apolitical	0.074	0.778	0.722
	left-wing liberal	-0.032	0.929	0.833
	right-wing conservative	-0.013	0.976	0.889
Sexual Orientation	bisexual	-0.075	0.795	0.667
	homosexual	-0.000	0.989	1.000
	other	0.101	0.696	0.611
	straight	-0.012	0.983	1.000